

LLMS, LRMS, AND REASONING: IN THE DOMAIN OF MEDICAL AI

DATASCI290: NEUROSymbolic AI, SPRING 2026
MARCH 9 2026

REASONING IN MEDICAL AI

■ Core questions:

- What does “reasoning” mean in medical AI?
- Do LLMs genuinely reason or simulate reasoning?
- What kinds of reasoning are required in clinical diagnosis?
- How can reasoning behavior in medical LLMs be evaluated?

■ References

- Sim, Shamus Zi Yang, and Tyrone Chen. (2025). Critique of impure reason: Unveiling the reasoning behaviour of medical large language models. eLife.
- Tu, Tao, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang et al. (2025). Towards conversational diagnostic artificial intelligence. Nature.
- Chen, Junying, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. (2025). Towards Medical Complex Reasoning with LLMs through Medical Verifiable Problems. Findings of ACL.

■ Focus: *Understanding reasoning processes, not just final answers.*



MODULES CLEANUP & REMAINING SESSIONS



LECTURES

- **Lecture 17: Medical Reasoning** Medical reasoning requires structured diagnostic thinking, verification of intermediate conclusions, and the ability to justify decisions. This lecture examines how modern LLM reasoning techniques - such as chain-of-thought, intermediate reasoning tokens, and verification steps, attempt to approximate these processes. We analyze the reasoning behavior of medical LLMs using the “Critique of Impure Reason” framework and examine how systems like AMIE incorporate structured evaluation loops for diagnostic dialogue. The lecture also introduces the concept of verifiable reasoning tasks designed to train LLMs to produce reasoning steps that can be checked.
 - Readings: Sim, Shamus Zi Yang, and Tyrone Chen. (2025). Critique of impure reason: Unveiling the reasoning behaviour of medical large language models. eLife. ; Tu, Tao, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang et al. (2025). Towards conversational diagnostic artificial intelligence. Nature.; Chen, Junying, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. (2025). Towards Medical Complex Reasoning with LLMs through Medical Verifiable Problems. Findings of ACL.
- **Lecture 18: Integrating Logic and LLMs** This lecture surveys emerging approaches for integrating symbolic logic with large language models to improve reasoning reliability. Topics include graph-based representations of reasoning chains, synthetic logic training corpora for improving logical consistency, and logic-augmented generation frameworks that combine symbolic reasoning structures with neural text generation. The goal is to combine the expressive capabilities of LLMs with the correctness guarantees of formal reasoning systems.
 - Readings: Alotaibi, F., Kulkarni, A., & Zhou, D. (2024). Graph of logic: Enhancing LLM reasoning with graphs and symbolic logic. IEEE BigData.; Morishita, T., Morio, G., Yamaguchi, A., & Sogawa, Y. (2024). Enhancing reasoning capabilities of LLMs via principled synthetic logic corpus. NeurIPS.; Xiao, Yilin, Chuang Zhou, Yujing Zhang, Qinggang Zhang, Su Dong, Shengyuan Chen, Chang Yang, and Xiao Huang. (2025). LAG: Logic-augmented generation from a Cartesian perspective.

LECTURES

- **Lecture 19: Neural Proposes, Symbolic Disposes: The Science of Diagnostic Reasoning for CareTrace** This lecture presents the architectural principle underlying CareTrace: neural models generate candidate interpretations and diagnostic hypotheses, while symbolic mechanisms determine which conclusions are justified. We examine diagnostic reasoning as a structured process involving abductive hypothesis generation, inductive pattern recognition, deductive rule enforcement, and sequential information gathering under uncertainty. The lecture focuses on defining the boundary between hypothesis proposal and decision authorization, ensuring that diagnostic conclusions are supported by evidence and consistent with clinical constraints. The resulting reasoning pipeline enables clinical AI systems whose outputs are verifiable, traceable, and safe for decision support.
- **Lecture 20: Putting It All Together: Designing Neurosymbolic Systems** This lecture synthesizes the major components developed throughout the course into a coherent neurosymbolic architecture. We revisit the progression from System-1 statistical models to System-2 symbolic reasoning, examining how knowledge graphs, KG-RAG pipelines, rule-based reasoning (Datalog), and logic-augmented LLM workflows can be combined in a single system. We also connect graph embeddings, large reasoning models, and clinical knowledge sources such as clinical practice guidelines to show how learning, knowledge representation, and formal reasoning interact. The lecture concludes with design principles for building trustworthy neurosymbolic systems that integrate neural proposal mechanisms with symbolic verification and structured knowledge.

LECTURES

- **Lecture 21: Building Multi-Agent Systems** Production neurosymbolic systems typically require coordination between multiple specialized components. This lecture introduces the architecture of multi-agent AI systems composed of perception agents, retrieval agents, reasoning agents, verification agents, and action agents. We examine LangGraph as a framework for orchestrating these agents, maintaining shared state, and integrating external systems such as knowledge graphs and reasoning engines into robust pipelines.
 - Readings LangGraph documentation
- **Lecture 22: Final Project Workshop on CareTrace** This project-focused lecture translates the full neurosymbolic architecture into the CareTrace clinical reasoning system. Students will define the knowledge graph schema, outline data ingestion pipelines, identify guideline rules and reasoning constraints, and design a multi-agent workflow for decision generation and verification. The lecture also discusses evaluation criteria including correctness, safety, traceability, and evidence-backed outputs.
- **Lecture 23: CareTrace Final Project Q&A** Open discussion session for the CareTrace project.

LECTURES

- **Lecture 24: Student Presentations Guidance**

- Readings

- Wang et al. (2023). Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023.
- Dhuliawala et al. (2024). Chain-of-Verification Reduces Hallucination in Large Language Models. ACL 2024 Findings.
- Ganguly et al. (2024). Proof of Thought: Neurosymbolic Program Synthesis Allows Robust and Interpretable Reasoning. NeurIPS 2024 Workshop on System-2 Reasoning.
- Besta et al. (2024). Graph of Thoughts: Solving elaborate problems with large language models. PVLDB 17(10), 2411–2425.
- Sun et al. (2024). Think-on-Graph: Deep and Responsible Reasoning of Large Language Model on Knowledge Graph. ICLR 2024.
- Hubert et al. (2025). Olympiad-level formal mathematical reasoning with reinforcement learning (AlphaProof). Nature.
- Trinh et al. (2024). AlphaGeometry: Solving olympiad geometry without human demonstrations. Nature 630, 1042–1049.
- Li et al. (2025). Proving Olympiad Inequalities by Synergizing LLMs and Symbolic Reasoning. ICLR 2025.
- Balcan, Maria-Florina, Avrim Blum, Zhiyuan Li, and Dravyansh Sharma. "On learning verifiers for chain-of-thought reasoning." arXiv preprint arXiv:2505.22650 (2025).
- Sikka, Varin, and Vishal Sikka. "Hallucination stations: On some basic limitations of transformer-based language models." arXiv preprint arXiv:2507.07505 (2025).
- Domingos, P. (2025). Tensor Logic: The Language of AI. arXiv preprint arXiv:2510.12269.

- **Lectures 25 & 26: Student Group Presentations**

- **Lectures 27 & 28: Final Project Presentations**



CONVERSATIONAL MEDICAL AI

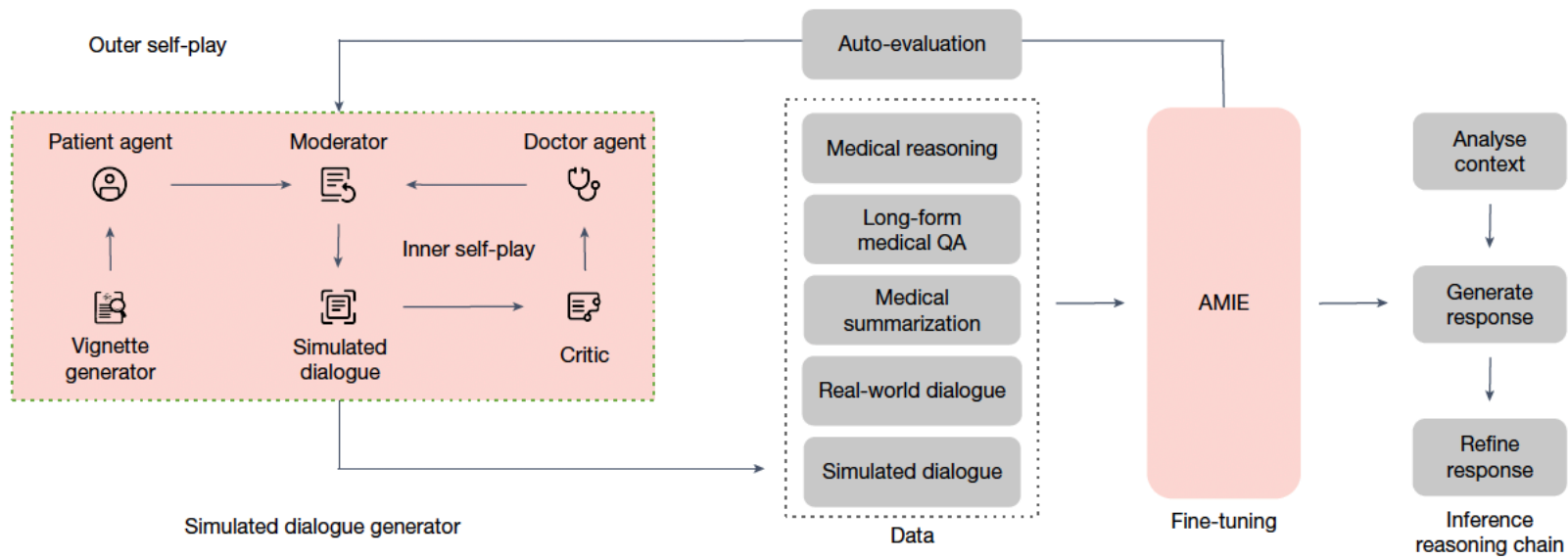


CONVERSATIONAL DIAGNOSTIC AI (AMIE)

- AMIE is an LLM optimized for clinical diagnostic dialogue.
 - Based on PalM-2
- Capabilities:
 - medical history-taking
 - diagnostic reasoning
 - patient interaction
- Training:
 - medical datasets
 - simulated doctor-patient dialogues
 - reasoning tasks
- Evaluation: Randomized OSCE-style (Objective Structured Clinical Exam) trial comparing AMIE and physicians.

AMIE SELF-PLAY TRAINING

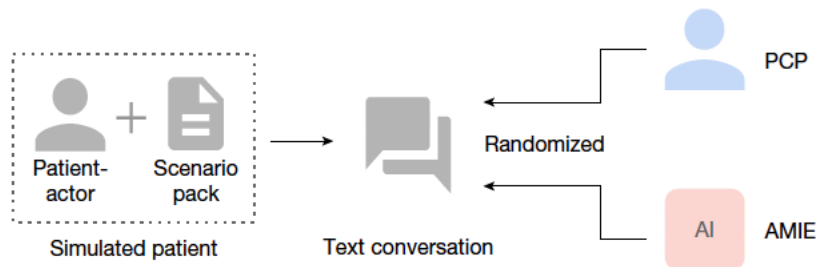
- AMIE training uses *simulated dialogue* generation.
- Two agents:
 - doctor agent
 - simulated patient agent
- Patient agent:
 - assigned medical case
 - symptoms and history
- Doctor agent:
 - asks questions
 - gathers information
 - generates differential diagnosis
- Dialogue becomes training data.



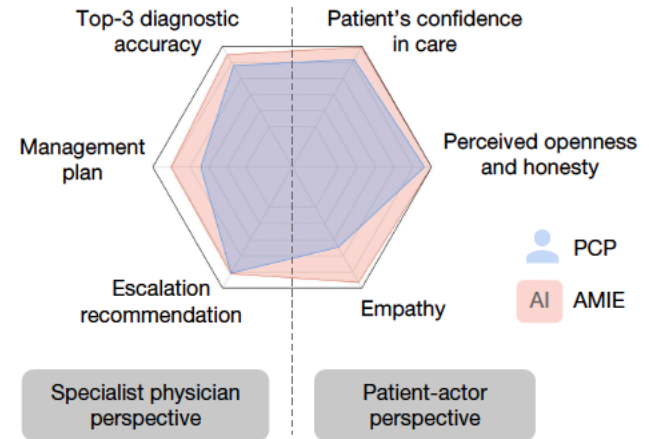
AMIE system design

AMIE EVALUATION STUDY

- Evaluation design:
 - 59 clinical scenarios
 - trained patient actors
 - 20 primary care physicians
- Each patient actor consults:
 - AMIE
 - a physician
- Evaluation axes:
 - diagnostic accuracy

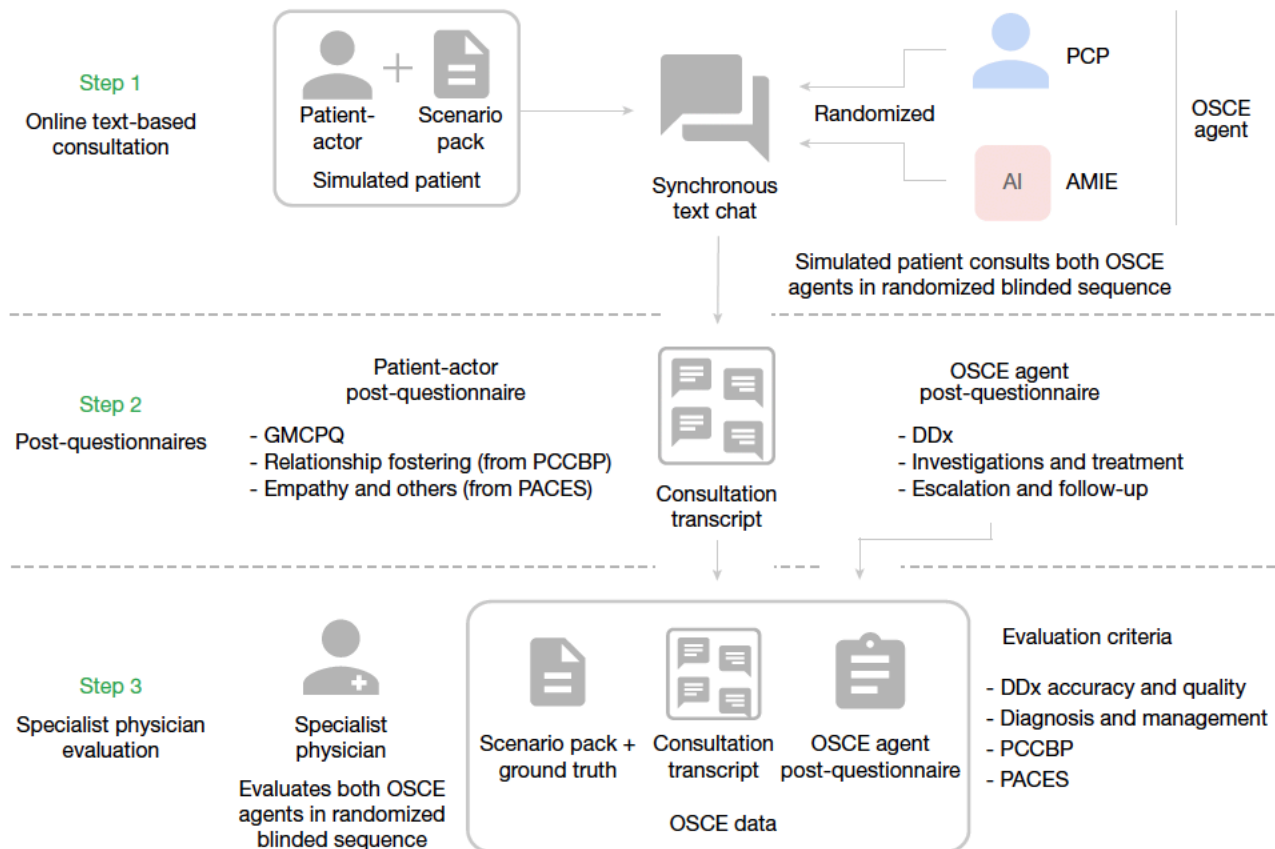


Randomized study design for remote OSCE

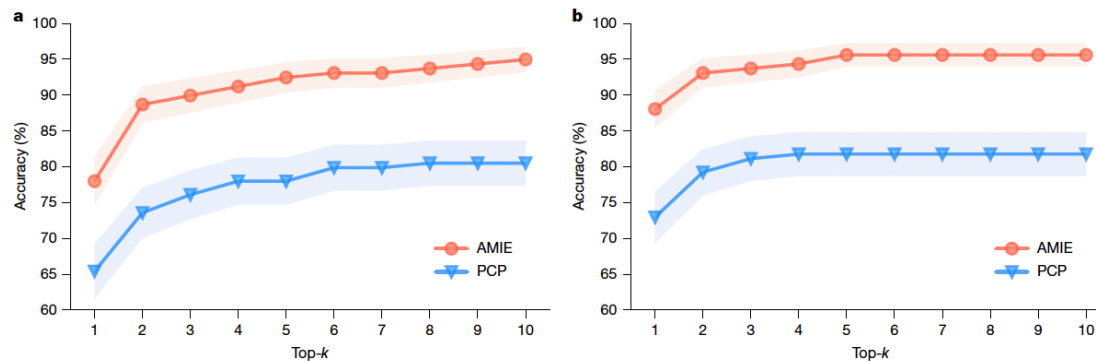


AMIE outperforms PCPs on multiple evaluation axes for diagnostic dialogue

AMIE EVALUATION

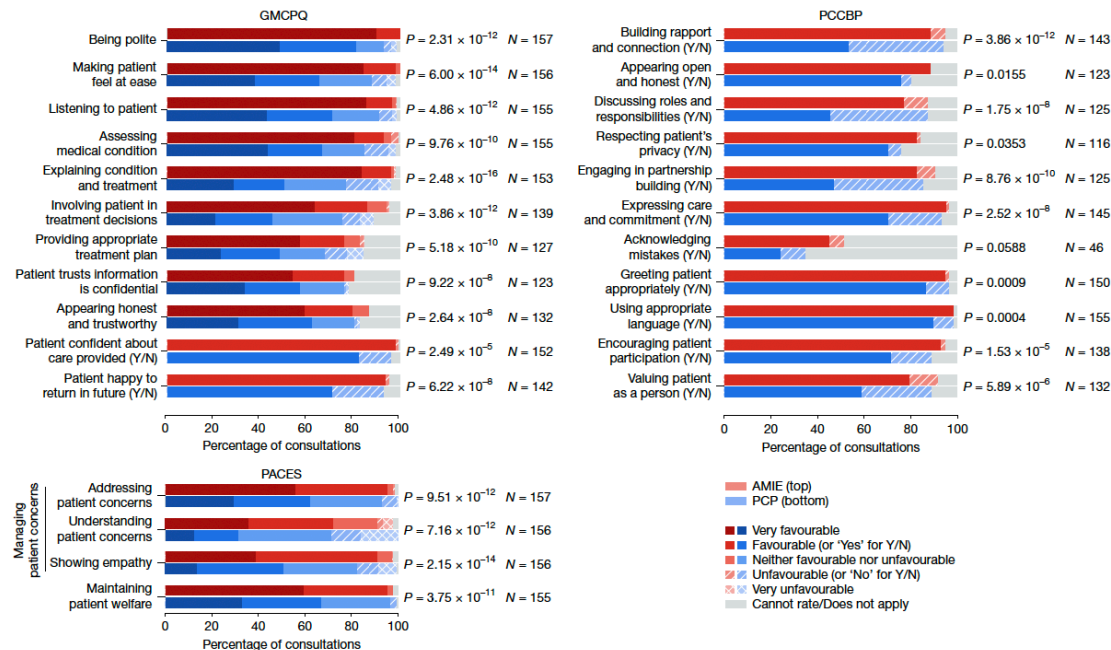


RESULTS: DIFFERENTIAL DIAGNOSIS



- In a blinded OSCE-style study with simulated patient consultations, **AMIE achieved higher differential diagnosis accuracy than PCPs** on specialist evaluation.
- The improvement held across **top-k differential diagnosis measures**, indicating more complete and accurate hypothesis sets.
- PCPs typically listed **fewer diagnoses in their differentials**, while AMIE produced broader candidate lists.
- Both AMIE and PCPs gathered **similar amounts of clinical information**, suggesting the advantage came mainly from **better interpretation of the evidence collected**.

RESULTS



- Patient actors rated AMIE **higher on most communication dimensions**, including clarity, empathy, and making the patient feel heard.
- Specialist physicians also preferred AMIE on **overall consultation quality** and several aspects of clinical reasoning and management discussion.
- Overall, AMIE was **preferred on the large majority of evaluation criteria**, while being at least non-inferior on the remainder.



CLINICAL REASONING



REASONING BEHAVIOUR IN MEDICAL LLMS

- Traditional evaluation focuses on final accuracy.
- But reasoning behavior asks:
 - How does the model arrive at its answer?
 - Medical LLM research rarely analyzes reasoning processes.
 - This motivates the “Critique of Impure Reason” study.

CORE THESIS OF CRITIQUE OF IMPURE REASON

- Key premise: Medical LLM research focuses excessively on outcome metrics:
 - accuracy
 - F1 score
 - exam performance
- These metrics do not reveal reasoning behavior.
- Position that reasoning transparency is essential for clinical trust.

CLINICAL REASONING IS ... DIFFERENT FROM QA

- Medical reasoning is fundamentally different from standard NLP tasks.
- Key properties:
 - Observations are incomplete and uncertain.
 - Evidence emerges over time through dialogue.
 - Multiple hypotheses must be maintained simultaneously.
 - Decisions are safety-critical.
- Example:
 - Patient complaint: chest pain
- Possible hypotheses:
 - myocardial infarction
 - pulmonary embolism
 - GERD
 - musculoskeletal injury
- Reasoning task: Generate, evaluate, and update competing hypotheses.

LOGICAL REASONING: DEFINITION

- Logical reasoning *derives conclusions from premises*.
 - Structure:
 - Premises → inference rules → conclusions
 - Example:
 - Premise 1: Fever often indicates infection.
 - Premise 2: Patient has fever and cough.
 - Conclusion: Respiratory infection is plausible.
- In medicine logical reasoning supports:
 - diagnostic inference
 - treatment decisions
 - clinical guidelines.

TYPES OF LOGICAL REASONING

- **1. Deductive reasoning**
 - General rule → specific case.
 - Example:
 - All bacterial pneumonia cases require antibiotics.
 - Patient has bacterial pneumonia.
 - Therefore antibiotics are indicated.
- **2. Inductive reasoning**
 - Observed cases → probabilistic generalization.
 - Example:
 - Most pneumonia patients show fever + cough.
 - Patient shows fever + cough.
 - Probability of pneumonia increases.
- **3. Abductive reasoning**
 - Observation → most plausible explanation.
 - Example:
 - Symptoms: fever + cough + shortness of breath.
 - Most plausible explanation: pneumonia.

ABDUCTIVE REASONING AND DIAGNOSIS

- Abduction *generates explanatory hypotheses*.
 - Observation:
 - fever
 - chest pain
 - cough
 - Possible explanations:
 - pneumonia
 - pulmonary embolism
 - COVID-19
- Abductive reasoning **proposes** hypotheses.
- Further evidence determines which hypothesis survives.

CAUSAL REASONING

- Causal reasoning *explains* mechanisms linking events.
 - Example:
 - Infection → inflammation → fever
- Causal knowledge helps physicians:
 - understand disease mechanisms
 - predict progression
 - choose appropriate therapy
- Causal reasoning differs from statistical correlation.

COUNTERFACTUAL REASONING

“WHAT-IF”

- Counterfactual reasoning evaluates alternative realities.
 - Example:
 - If antibiotics had been given earlier, would sepsis have occurred?
- Clinical applications:
 - evaluating treatment decisions
 - understanding disease progression
 - policy and treatment design.

Type of reasoning	Definition	General example	Medical example
Abductive	Inferring the most likely explanation for observed data or evidence.	Ali, Muthu, and Ah Hock breathe oxygen. Therefore, Ali, Muthu, and Ah Hock are likely human.	A patient has increased intracranial pressure, blurred vision and nausea. Therefore, the patient may have a brain aneurysm or ischaemic stroke.
Deductive	Reasoning from a set of premises to reach a certain conclusion.	All humans breathe oxygen. Rentap is human. Therefore, Rentap breathes oxygen.	A patient has increased intracranial pressure, blurred vision and nausea. A CT scan shows no bleeding or swelling. Therefore, the patient does not have a brain aneurysm.
Inductive	Inferring general principles based on specific observations.	All humans that I have seen breathe oxygen. Therefore, Rentap probably breathes oxygen.	A patient has increased intracranial pressure, blurred vision and nausea. A CT scan shows no bleeding or swelling. Therefore, the patient probably has an ischaemic stroke.
Symbolic*	The abstraction of a system into its component parts, which enables a more direct application of mathematics.	Rule: If an organism breathes oxygen and nitrogen, and exhales carbon dioxide → likely human. Observation: Ali, Muthu, Ah Hock, and Rentap exhibit this respiratory pattern. Conclusion: Therefore, they are probably human.	Rule 1: If a patient presents with increased intracranial pressure (ICP), blurred vision, and nausea → infer high intracranial pathology. Observation 1: The patient shows increased ICP, blurred vision, and nausea. Rule 2: If CT scan shows no bleeding or swelling → rule out haemorrhagic causes. Observation 2: CT scan reveals no evidence of bleeding or swelling. Rule 3: If high ICP and haemorrhage is ruled out → suspect ischaemic stroke. Conclusion: Therefore, the patient most likely has an ischaemic stroke.
Causal/ Counterfactual	Establishing a cause-and-effect relationship between events.	Ali, Muthu, Ah Hock, and Rentap exhibit this respiratory pattern.	A blood clot probably caused blockage in the brain leading to the stroke.

Method name	Base architecture/ method	Reasoning improvement strategy	Type of Reasoning	Advantages	Disadvantages	Dataset	Github
<i>Savage et al., 2024</i>	GPT-3.5; GPT-4.0	Chain-of-thought (diagnostic reasoning)	Deductive	Easy to implement	Scope is limited to GPT models, focusing exclusively on English medical questions	Modified MedQA USMLE; NEJM (New England Journal of Medicine) case series	
<i>Kwon et al., 2024</i>	GPT-4.0; OPT; LLaMA-2; 3D ResNet	Chain-of-thought; knowledge distillation (via SFT)	Deductive	Lightweight and practical to use	Tight scope to limited disease conditions	Alzheimer's Disease Neuroimaging Initiative (ADNI); Australian Imaging Biomarkers and Lifestyle Study of Ageing (AIBL)	https://github.com/ktio89/ClinicalCoT
MEDDM Binbin et al <i>Li et al., 2023</i>	GPT	Chain-of-thought; clinical decision trees	Deductive	Adaptable to different systems	Heavy data collection to generate clinical guidance trees	Medical books, treatment guidelines, and other medical literature	
DRHOUSE <i>Yang et al., 2024a</i>	GPT-3.5; GPT-4.0; LLaMA-3 70b; HuatuoGPT-II; MEDDM	Chain-of-thought; clinical decision trees	Deductive	Objective sensor measurement	Available datasets are currently limited	MedDG; KaMed; DialMed	
DR. KNOWS <i>Gao et al., 2023b</i>	Vanilla T5; Flan T5; ClinicalT5; GPT	Chain-of-thought; extracted explainable diagnostic pathway	Deductive; neurosymbolic	Hybrid method improves accuracy; provides explainable diagnostic pathways	Particularly fragile to missing data	MIMIC-III; In-house EHR	
TEMED-LLM <i>Bisercic et al., 2023</i>	text-davinci-003; GPT-3.5; logistic regression; decision tree; XGBoost	Few-shot learning, tabular ML modelling; Neurosymbolic	Deductive	End-to-end interpretability, from data extraction to ML analysis	Requires human experts	EHR dataset (kaggle; see referenced publication for details)	
EHRAgent <i>Shi et al., 2024</i>	GPT-4	Autonomous code generation and execution for multi-tabular reasoning in EHRs	Deductive	Facilitates automated solutions in complex medical scenarios	Non-deterministic; limited generalisability	MIMIC-III; eICU; TREQS	https://github.com/wshi83/EhrAgent ; https://wshi83.github.io/EHR-Agent-page
AMIE <i>Tu et al., 2024</i>	PaLM 2	Reinforcement learning	Deductive	Effectively handles noisy and ambiguous real-world medical dialogues	Computationally expensive and resource-intensive; simulated data may not fully capture real-world clinical nuances	MedQA; HealthSearchQA; LiveQA; Medication QA in MultiMedBench, MIMIC-III	
ArgMed-Agents <i>Hong et al., 2024</i>	GPT-3.5-turbo; GPT-4	Chain-of-Thought; symbolic reasoning; neurosymbolic	Deductive	Training-free enhancement; explainability matches fully transparent, knowledge-based	Artificially restricted responses that do not match real-world cases	MedQA; PubMedQA	

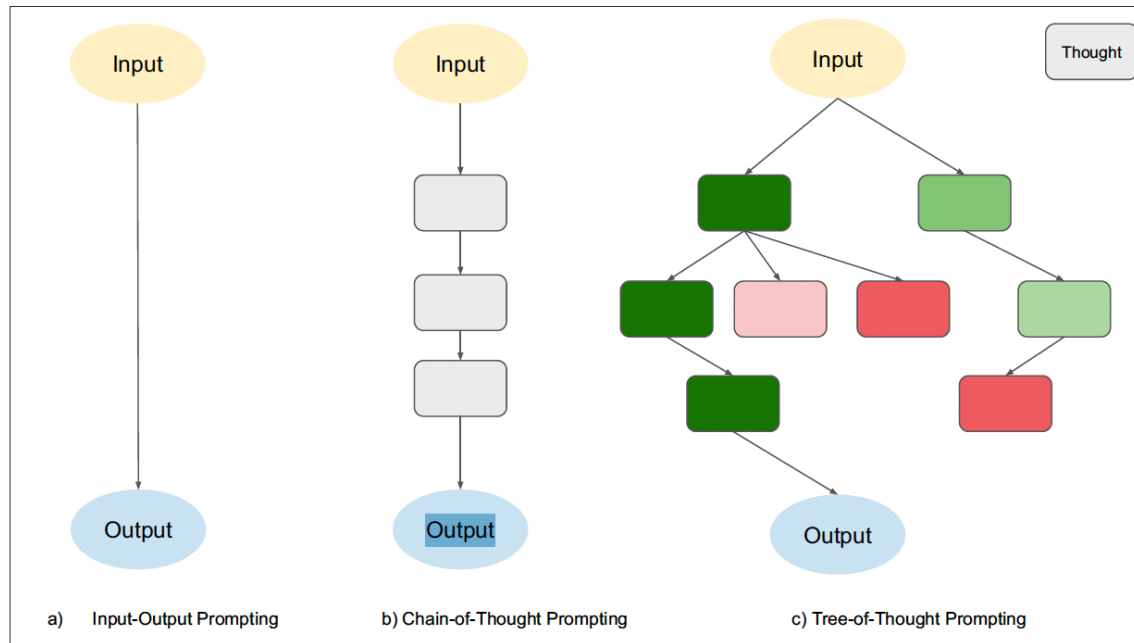
IMPURE REASONING

- Impure reasoning refers to reasoning traces that appear logical but are not causally responsible for the model's answer.
- LLMs may:
 - generate explanations after producing answers
 - produce plausible reasoning narratives
 - mimic clinical reasoning patterns
- These explanations may not reflect true inference.

REASONING NARRATIVES VS REAL REASONING

- True reasoning: observations $\rightarrow$ inference $\rightarrow$ conclusion
- LLM behavior: observation $\rightarrow$ predicted answer $\rightarrow$ generated explanation
- This creates an illusion of reasoning.

PROMPTING STRATEGIES FOR REASONING



- Prompting methods attempt to improve reasoning transparency.
 - Chain-of-Thought (CoT): Model generates step-by-step reasoning.
 - Tree-of-Thought (ToT): Model explores multiple reasoning branches.
- Goal: Encourage structured reasoning instead of single-step answers.

CHAIN-OF-THOUGHT REASONING

- Chain-of-Thought example:
- Patient symptoms:
 - fever + cough
- Model reasoning:
 - 1. Fever indicates infection.
 - 2. Cough suggests respiratory involvement.
 - 3. Pneumonia becomes plausible.
- However reasoning steps may be generated post-hoc.

TREE-OF-THOUGHT REASONING

- Tree-of-Thought explores multiple reasoning paths.
- Example differential reasoning:
 - Chest pain
 - ⊢ cardiac cause
 - ⊢ pulmonary cause
 - ⊢ gastrointestinal cause
 - ⊢ musculoskeletal cause
- Each branch is explored before choosing a conclusion.

AGENT-BASED REASONING

- Agent architectures assign different reasoning roles to multiple models.
- Example agents:
 - diagnostic reasoning agent
 - knowledge retrieval agent
 - safety verification agent
- Agents collaborate to evaluate clinical scenarios.

LEARNING TO REASON IN LLMS

- Training approaches for reasoning models:
 - Supervised Fine-Tuning (SFT): Model trained on examples of reasoning steps.
 - Reinforcement Learning with Human Feedback (RLHF): Humans evaluate model outputs and guide training.
- Goal: Align model behavior with expert reasoning.

INFERENCE-TIME REASONING SCALING

- Models can also reason more deeply at inference time.
- Techniques:
 - generating multiple reasoning paths
 - self-evaluation of reasoning chains
 - selecting best reasoning path
- This is sometimes called test-time scaling.

REASONING FAILURE MODE: PREMATURE CLOSURE

- Premature closure occurs when reasoning stops too early.
- Example:
 - Symptoms:
 - chest pain
 - nausea
 - Model concludes GERD without considering myocardial infarction.
- Danger: Serious diagnoses may be ignored.

REASONING FAILURE MODE: HALLUCINATED EVIDENCE

- LLMs sometimes fabricate evidence during reasoning.
- Example:
 - Patient statement:
 - “I have abdominal pain.”
 - Model reasoning:
 - “Pain after meals suggests gallstones.”
- But meals were never mentioned !
- This is hallucinated reasoning evidence.

REASONING EVALUATION METHODS

- Propose analyzing reasoning behavior using multiple methods.
 - reasoning trace inspection
 - process supervision
 - logical consistency checks
 - counterfactual reasoning tests
 - adversarial diagnostic scenarios
- Goal: Evaluate reasoning quality, not just answers.

ADDITIONAL REASONING EVALUATION DIMENSIONS

Table 3. Comparison of reasoning evaluation paradigms in medical LLMs.

Evaluation paradigm	Conceptual focus	Typical implementation and metrics
Conclusion-based	Assesses correctness of the final answer only, without inspecting the reasoning path.	Automated scoring on Q-&-A benchmarks (e.g. MedQA, MedMCQA). Metrics: Accuracy, exact-match, F1. Fast, reproducible, but offers only high-level insight.
Rationale-based	Evaluates the logic chain or narrative explanation produced by the model. Focuses on coherence, validity, and completeness of reasoning traces.	Manual expert review or rubric-based grading of CoT. Automated graph checks (e.g. DAG similarity, causal-direction tests). Metrics: Bayesian Dirichlet score, Normalised Hamming Distance.
Mechanistic	Probes low-level internal signals to answer “ <i>why did the model arrive here?</i> ”. Targets feature attribution and internal attention contributions.	Explainable-AI toolkits (Integrated Gradients, SHAP, attention rollout). Outputs saliency maps or keyword heat-maps for clinician inspection.
Interactive	Treats evaluation as a dialogue or game ; dynamically stresses the model in real time. Explores the <i>response space</i> by challenging, re-prompting, or role-playing.	Game-theoretic tasks (e.g. debate, self-play). Rich insights but lower reproducibility; requires human-in-the-loop or scripted agents.

- Researchers analyze:
 - mechanistic reasoning
 - interactive reasoning in dialogue
 - rationale generation
 - logical inference structure

IMPLICATIONS FOR MEDICAL AI

- The critique concludes: LLMs alone cannot guarantee reliable reasoning.
- Statistical language models:
 - capture correlations
 - generate explanations
 - lack explicit causal reasoning structures
- Medical AI therefore requires structured reasoning systems.

NEUROSymbOLIC MEDICAL AI

- Neurosymbolic systems combine:
 - Neural models:
 - language understanding
 - hypothesis generation
 - Symbolic models:
 - medical knowledge graphs
 - diagnostic rules
 - constraint verification
- This architecture improves reasoning transparency.



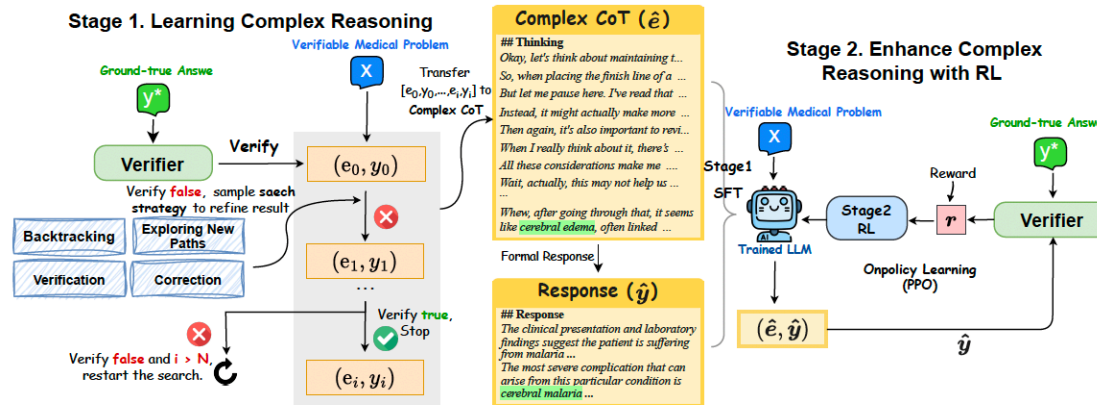
HUATUO-GPT



HUATUOGPT-O1

- Goal: bring o1-style long-chain medical reasoning into the medical domain, where strong reasoning is needed but verification is much harder than in math.
- HuaTuoGPT-o1
 - Core idea: convert difficult closed-set medical exam questions into 40K verifiable medical problems with open-ended answers and unique ground truth, so that reasoning outcomes can actually be checked.
 - Model family: HuatuoGPT-o1-8B and HuatuoGPT-o1-70B, built on Llama-3.1-8B-Instruct and Llama-3.1-70B-Instruct respectively.
- HuaTuoGPT-o1
 - Training data source: mainly MedQA-USMLE and MedMCQA training sets, filtered and reformatted; they also add some original exam-format data and a small amount of non-medical verifiable data from MMLU-Pro to improve generalization.
 - Main claim: with only 40K verifiable medical problems, they can train a medical LLM that performs strong complex reasoning, not just medical recall.

RESULTS



- **Stage 1:** search for correct reasoning trajectories. The model proposes an initial CoT; if the answer is wrong, it refines reasoning using one of four strategies: Backtracking, Exploring New Paths, Verification, and Correction. Successful searches are reformatted into a smoother Complex CoT and used for supervised fine-tuning.
- **Stage 2:** reinforcement learning with verifier rewards. After SFT, they use PPO so the model improves its long reasoning traces using sparse reward from the verifier.
- **Verification:** they use GPT-4o as the medical verifier, checking whether the final answer matches the ground-truth answer; on 200 manually checked samples, verifier accuracy was about 96.5% in Stage 1 and 94.5% in Stage 2, much better than exact match. They also show a fine-tuned 8B verifier can exceed 90% accuracy.
- **HuaTuoGPT-o1**
 - Benchmarks: evaluated on MedQA, MedMCQA, PubMedQA, MMLU-Pro medical tracks, and GPQA medical/biology tracks.
- **Key result:** the 8B model improves by about 8 points over Llama-3.1-8B-Instruct, and the 70B model outperforms open-source general and medical baselines; RL gives a further boost over SFT alone.